SKYWORTH

SWH
6/8/10/12/15kW
Three Phase Hybrid Inverter (HV)



MODEL

SWH6000H-T1-G2 | SWH8000H-T1-G2

SWH010KH-T1-G2 | SWH012KH-T1-G2

SWH015KH-T1-G2

Flexible & Adaptable Applications
- Smart load control with dry contacts
- Smart home integration with multi-protocol communications

Friendly & Thoughtful Design
- Plug & Play installations
- Elegant and compact design

Superb Safety & Reliability
- AI-driven AFCI 3.0[1]
- IP66 ingress protection
- Type II SPD on DC & AC sides

Higher Power Generation
- Unbalanced output up to 150%[2]
- Up to 160% PV input oversizing
- Parallel connection capability for increased output power

1: Optional functions or devices are purchased separately.
2: Only for the SWH6000H-T1-G2, SWH8000H-T1-G2, SWH010KH-T1-G2 models.

Model	SWH6000H-T1-G2	SWH8000H-T1-G2	SWH010KH-T1-G2	SWH012KH-T1-G2	SWH015KH-T1-G2
Battery Input Data					
Battery Type*1	Li-Ion				
Nominal Battery Voltage (V)	500				
Battery Voltage Range (V)	150 ~ 720				
Start-up Voltage (V)	150				
Number of Battery Input	1				
Max. Continuous Charging Current (A)	30	30	40	40	40
Max. Continuous Discharging Current (A)	30	30	40	40	40
Max. Charging Power (W)	9000	12000	15000	18000	24000
Max. Discharging Power (W)	6600	8800	11000	13200	16500
PV String Input Data					
Max. Input Power (W)*2	9600	12800	16000	19200	24000
Max. Input Voltage (V)*3*4	1000				
MPPT Operating Voltage Range (V)*5	120 ~ 850				
Start-up Voltage (V)	150				
Nominal Input Voltage (V)	620				
Max. Input Current per MPPT (A)	16				
Max. Short Circuit Current per MPPT (A)	24				
Number of MPP Trackers	2	2	3	3	3
Number of Strings per MPPT	1				
AC Output Data (On-grid)					
Nominal Output Power (W)	6000	8000	10000	12000	15000
Nominal Apparent Power Output to Utility Grid (VA)	6000	8000	10000	12000	15000
Max. Apparent Power Output to Utility Grid (VA)*6	6000	8000	10000	12000	15000
Max. Apparent Power from Utility Grid (VA)	12000	16000	20000	20000	20000
Nominal Output Voltage (V)	400 / 380, 3L / N / PE				
Output Voltage Range (V)*7	170 ~ 290				
Nominal AC Grid Frequency (Hz)	50 / 60				
AC Grid Frequency Range (Hz)	45 ~ 65				
Max. AC Current Output to Utility Grid (A)*8	8.7	11.6	14.5	17.4	21.7
Max. AC Current From Utility Grid (A)	15.7	21.0	26.1	26.1	26.1
Power Factor	0.8 leading ~ 0.8 lagging				
Max. Total Harmonic Distortion	<3%				
AC Output Data (Back-up)					
Back-up Nominal Apparent Power (VA)	6000	8000	10000	12000	15000
Max. Output Apparent Power without Grid (VA)*10	6000 (12000 @60sec)	8000 (16000 @60sec)	10000 (18000 @60sec)	12000 (18000 @60sec)	15000 (18000 @60sec)
Max. Output Apparent Power with Grid (VA)	6000	8000	10000	12000	15000
Max. Output Current (A)	13.0 (17.4 @60sec)	17.4 (23.3 @60sec)	21.7 (26.1 @60sec)	21.7 (26.1 @60sec)	21.7 (26.1 @60sec)
Nominal Output Voltage (V)	400 / 380				
Nominal Output Frequency (Hz)	50 / 60				
Output THDv (@Linear Load)	<3%				
Efficiency					
Max. Efficiency	98.0%	98.0%	98.2%	98.2%	98.2%
European Efficiency	97.2%	97.2%	97.5%	97.5%	97.5%
Max. Battery to AC Efficiency	97.2%	97.5%	97.5%	97.5%	97.5%
MPPT Efficiency	99.5%				
Protection					
PV Insulation Resistance Detection	Integrated				
PV AFCI3.0	Optional				
Residual Current Monitoring	Integrated				
PV Reverse Polarity Protection	Integrated				
Battery Reverse Polarity Protection	Integrated				
Anti-islanding Protection	Integrated				
AC Overcurrent Protection	Integrated				
AC Short Circuit Protection	Integrated				
AC Overvoltage Protection	Integrated				
DC Switch	Integrated				
DC Surge Protection	Type II				
AC Surge Protection	Type II				
Remote Shutdown	Integrated				
General Data					
Operating Temperature Range (°C)	-35 ~ +60				
Relative Humidity	0 ~ 100%				
Max. Operating Altitude (m)	4000				
Cooling Method	Natural Convection				
User Interface	LED, WLAN + APP				
Communication with BMS	RS485, CAN				
Communication with Meter	RS485				
Communication with Portal	WiFi + LAN + Bluetooth				
Weight (kg)	23	23	25	25	25
Dimension (W × H × D mm)	496 × 460 × 221				
Topology	Non-isolated				
Ingress Protection Rating	IP66				
Mounting Method	Wall Mounted				

*1: The Li-Ion battery usually contain two mainstream type: LFP and Ternary Lithium battery.
*2: Max. input power, not continuous for 1.6*normal power. Besides, in Australia, for most of the PV modules, the max. input power can reach 2*Pn, for example, the max. input power of SWH6000H-T1-G2 can reach 12000W.
*3: For 1000V system, the maximum operating voltage is 950V.
*4: When the input voltage ranges from 975V to 1000V, the inverter will enter the standby mode, and the voltage returns to 975V to enter the normal operation state.
*5: Please refer to the user manual for the MPPT Voltage Range at Nominal Power.
*6: According to the local grid regulation.
*7: Output Voltage Range: phase voltage.
*8: When the three-unbalance function is activated, the Max. AC Current Output to the on-grid load can reach 13A, 17.4A, 21.7A, 21.7A and 21.7A respectively.
*9: If the inverter is installed with the 3x25A AC breaker, it is suggested that the consumption and feed-in AC power should be less than 11040W (0.8×0.8×25×230×3), and this limitation can be set by App.
*10: Can be reached only if PV and battery power is enough.
*: Please visit SKYWORTH website for the latest certificates.

SKYWORTH
00751.HK
Tel: 400-063-8989 Email: solar.overseas@skyworth.com www.skyworth-pv.com
All Rights Reserved @Shenzhen Skyworth Photovoltaic Technology Co.,Ltd. Please refer to the latest version for any updates.